Kefei Zhu

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Education

Beijing University of Posts and Telecommunications *Sep 2021 - Jun 2024*
M.Sc. in Management Science and Engineering (top 3%)

China Jiliang University *Sep 2017 - Jun 2021*
B.Eng. in Industrial Engineering (top 3%)

Research Experience

Peking University – PrimeBot Joint Lab *Jan 2026 - Present*
Research Intern — Advisor: ***Hao Dong*** *Beijing, China*
Researching fine-grained dexterous manipulation via human video grounding and world model.

Peking University – PsiBot Joint Lab *Sep 2024 - Jan 2026*
Research Intern — Advisors: ***Yuanpei Chen, Yaodong Yang*** *Beijing, China*
Researching dexterous manipulation data flywheel and building continual learning systems for robotics.

Institute of Automation, Chinese Academy of Sciences *Sep 2023 - Aug 2024*
Research Intern — Advisor: ***Dongbin Zhao*** *Beijing, China*
Conducted research on imitation learning and diffusion model for autonomous driving.

Industry Experience

Zeekr Auto *May 2023 - Aug 2023*
Algorithm Intern — Mentor: ***Tao Guo*** *Shanghai, China*
Developed motion planning algorithms for real-vehicle autonomous driving systems.

Research Projects

Data Generation Framework for Dexterous Manipulation
Project lead and first author arXiv Project Website Media

- Leading the development of **DexFlyWheel**, a scalable and self-improving data generation framework for dexterous manipulation that forms a closed-loop flywheel by integrating imitation learning with reinforcement learning to iteratively improve policy quality and data diversity.

Continual Fine-Tuning Pretrained Robot Policies via Reinforcement Learning
Project lead and first author

- Proposed a continual fine-tuning framework for pretrained robot policies using residual reinforcement learning and model merging to enable continual adaptation across sequential tasks.

Publications

Kefei Zhu, Fengshuo Bai, YuanHao Xiang, Yishuai Cai, Xinglin Chen, Ruochong Li, Xingtao Wang, Hao Dong, Yaodong Yang, Xiaopeng Fan, Yuanpei Chen†, "DexFlyWheel: A Scalable and Self-improving Data Generation Framework for Dexterous Manipulation",**NeurIPS 2025 Spotlight (Acceptance rate:3.1%)**.

Pengxuan Yang, Yupeng Zheng, Qichao Zhang†, **Kefei Zhu**, Zebin Xing, Qiao Lin, Yun-Fu Liu, Zhiguo Su, Dongbin Zhao, "UncAD: Towards Safe End-to-end Autonomous Driving via Online Map Uncertainty", **ICRA 2025**, **Accepted**. arXiv.

Xinglin Chen, Yishuai Cai, Minglong Li, Wenjing Yang, **Kefei Zhu**, Fengshuo Bai, Yaodong Yang, Yuanpei Chen†, "ToS: Tree-of-Skill Reinforcement Learning driven by Behavior Tree for Long-horizon Manipulation", **in Submission**.

Miscellaneous

Languages: English (IELTS 7.0), Chinese (Native)
Academic Service: Reviewer for ICRA 2026, IEEE Student Member 2025-Present
Technical Skills: Python, C/C++, ROS2, PyTorch, LibTorch, Git, OmniGibson

Honors

Graduate First-Class Scholarship	*Jun 2023*
Graduate First-Class Scholarship	*Oct 2022*
Graduate First-Class Scholarship	*Sep 2021*
Outstanding Undergraduate Graduate of Zhejiang Province (3%)	*Jun 2021*
First Prize in Advanced Mathematics Competition (Engineering Category)	*Dec 2020*
Zhejiang Provincial Government Scholarship (5%)	*Jun 2020*
World Internet Conference Volunteer Service	*Oct 2019*